

Ahmed GHALEB

Al-Muneer Hall Reservations by Ahmed Ghaleb- PSM 1-25-32

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CHAPTER 3

SYSTEM DEVELOPMENT METHODOLOGY

3.1 Introduction

The development of the Al-Muneer Online Portal having requirements and scope specified needs to follow a systematic and structured methodological approach so as to ensure that all objectives are accomplished in an efficient and effective manner. Right from the first concept and then through the planning and design phases associated with this PSM1 stage, the methodology that will be followed by the project should be such that it will take it through all of the entire life cycle before taking it further towards development and deployment in the PSM2 stage. System development methodology to be followed for the development of the Al-Muneer Online Portal, its various phases and methodology implementation for the accomplishment of objectives of the project have been discussed in this chapter. A suitable methodology for a project of this nature where requirements are already known and structured approach is needed is Waterfall methodology.

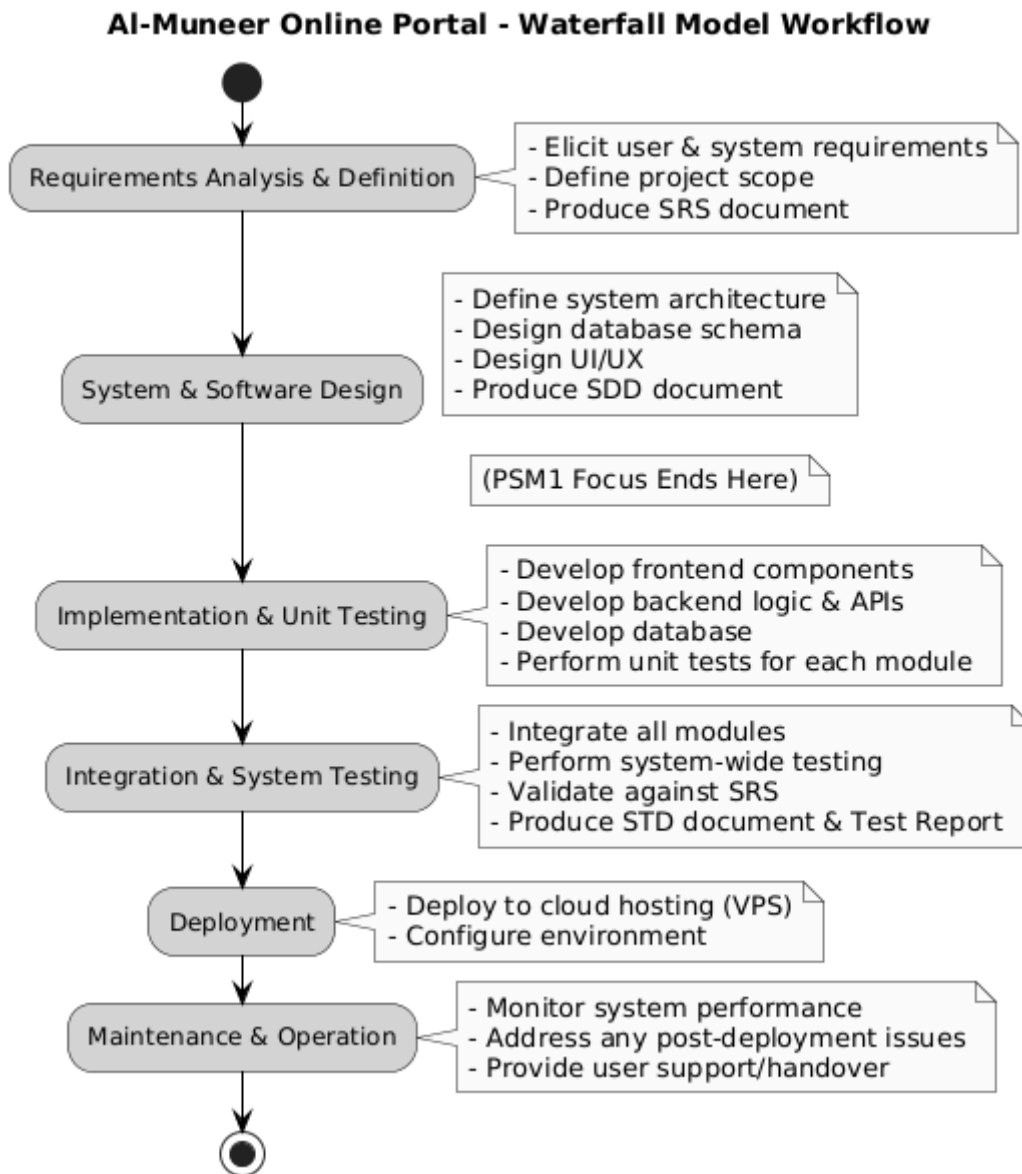
3.2 Methodology Choice and Justification

Given the linearity and sequential process of software development using the Waterfall model, this model is chosen for the development of the Al-Muneer Online Portal project. This model requires that there be segmentation of the project life cycle into phases and completion of one phase prior to moving on to the next. Given that there has been some preliminary coordination and clarification regarding the proposal, the Al-Muneer Online Portal project is well positioned to benefit from such a rigid model since it is suitable for projects whose requirements have been specified from the onset of the project. The distinct scope of functionalities that include an interactive calendar, prices displayed, uploading payment proofs, feedback, ease of reporting and notification make this model appropriate.

Life cycles in the waterfall approach should typically be used in cases where predictability and levels of documentation are key criteria. The clear identification of the stages in the Waterfall approach ensures established indicators to measure progress and outcomes such as SDD and SRS in terms of FYP student project with defined academic milestones (Progress 1, Progress 2, Final Report). Agile methods allow flexible approaches to project execution especially when there is a change in requirements; however, the Al-Muneer Online Portal is better off with the Waterfall approach in light of its strong emphasis on early planning and designing, thus

ensuring lower uncertainty while shifting from the planning and designing phase in PSM1 to developing phase in PSM2. Such an approach will ensure the certainty that the developing process in PSM2 can proceed upon a solid base that has been agreed on by all the parties involved in the project.

3.2.1 Waterfall Model Process Diagram



Figur

e 3.1: Waterfall Model Workflow for the AI-Muneer Online Portal

3.3 Phases of the Chosen Methodology

The development life cycle of Al-Muneer Online Portal based on the Waterfall model has the different phases. Each phase has its own goals and products to be delivered and they lead in an orderly way to the final product.

1. **Requirements Analysis and Definition**

This stage is very important to understand and capture the special needs of the Al-Muneer Online Portal. Detailed discussions with stakeholder (Mr. Almunajid) are included for capturing all functional and non-functional requirements. The involved activities are determination of users' roles (visitor and admin), listing of functionalities which include presentation of venue details, media gallery, availability calendar, ability to customize pricing, submission of booking requests, uploading of payments, providing feedback, administration of content by the admin, provision of payment status updates, review of feedbacks, generation of basic reports and setting up of notifications. The primary output of this phase is the Software Requirement Specification (SRS). This stage was one of the main activities in the PSM1 stage.

2. **System and Software Design**

Once requirements are defined and documented in SRS document, design phase begins. In this phase total system architecture of Al-Muneer Online Portal is designed. It includes designing the database schema (MySQL/PostgreSQL), designing the backend (Spring Boot services/APIs) and designing the UI and UX for the front end for visitors and the back end for the administration. Decisions about module segregation, flow of data, and security measures are determined during this phase. The main output of this phase is the System Design Document (SDD), which acts as a guideline for the development phase (student developer). This is an essential part of the PSM1 process, especially before Progress 2.

3. **Implementation and Unit Testing**

This is the point where real coding and development commence, and most of it will be done in PSM2. As per the SDD, the Al-Muneer Online Portal will be developed. The web application will be built in one integrated project. The backend logic will be handled using Spring Boot (Java), and will also act as the UI for the portal. The View layer will be designed using server-side templating engine (Thymeleaf) together with HTML, CSS and JavaScript. The database will be set up and populated with some structures. Each component will be tested as it is being developed in order to confirm its functionality.

4. Integration and System Testing

Once the units are completed and tested individually, they are integrated to form a complete system. The complete system then goes through system testing to ensure that all components work well together as a system and that the entire system meets the requirements stated in the SRS. This includes functional testing, usability testing, basic performance testing, and security testing. An STD will be used for this purpose, and a test report will be compiled from it. This stage is also meant to take place in PSM2.

5. Deployment

The migration of Al-Muneer Online Portal to a production-like environment will only take place once system testing is completed and found to be stable, adhering to all the stipulated requirements. Migration will take place on Cloud VPS. This phase will encompass the setting up of hosting environment, deployment of the application code and database, and finally performing testing so that the portal can become operational and accessible by users. It is likely to be completed during the latter part of PSM2.

6. Maintenance and Operation

Post deployment comes the phase of maintenance. This phase involves constant checking of the Al-Muneer Online Portal for any problem or bug that needs fixing or updating. In the FYP scenario, the phase can involve a process of observation, sorting out any problem after the deployment phase, and preparing documents in order to hand over the system to the stakeholder. Even though long term maintenance is not included in an FYP, it's important to ensure that the system is stable before handing it over.

3.3.1 Gantt Chart for PSM1

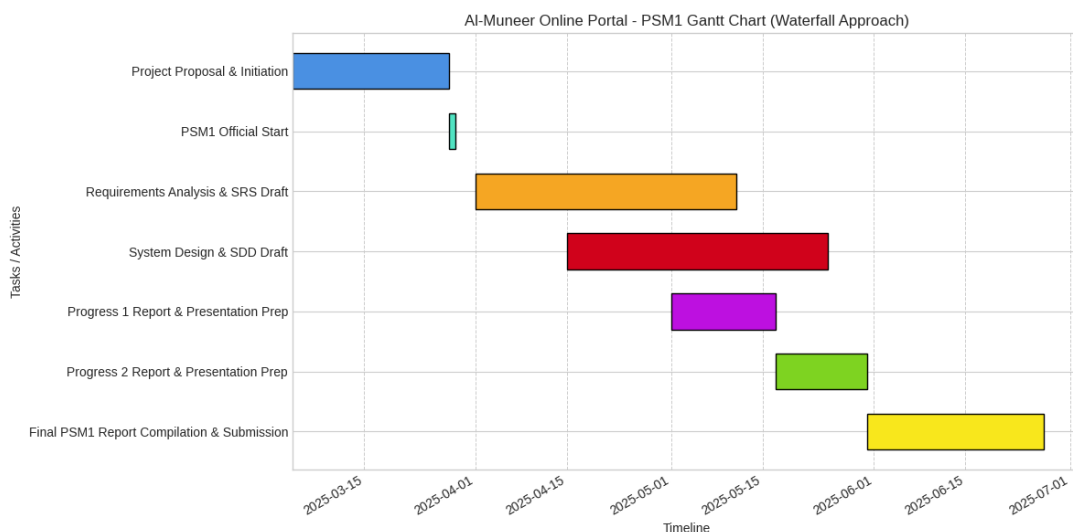


Figure 3.2: Gantt Chart for PSM1 Activities

3.3.2 Gantt Chart for PSM2

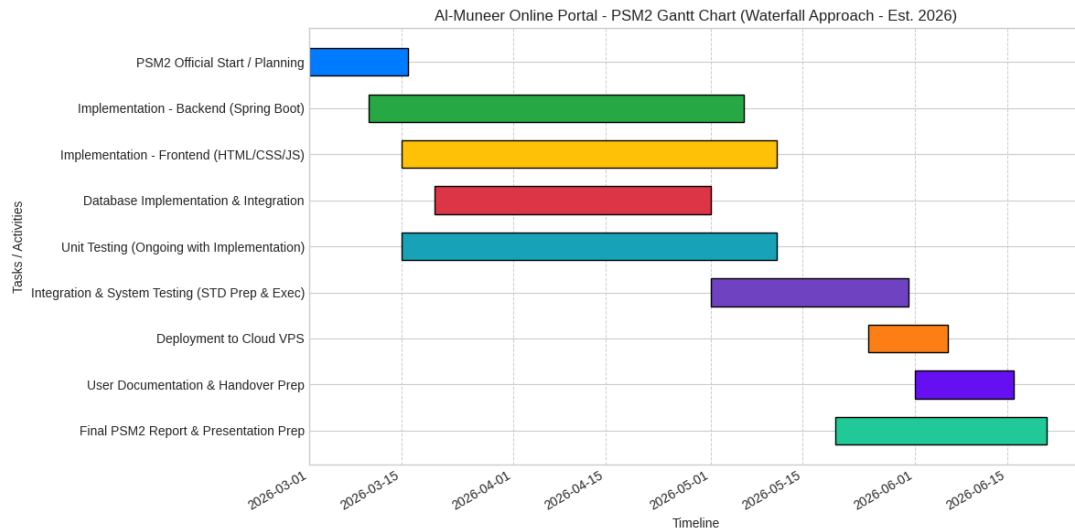


Figure 3.3: Proposed Gantt Chart for PSM2 Activities

3.4 System Requirements Analysis: Hardware and Software

The successful development and deployment of the AI-Muneer Online Portal will require certain hardware and software resources. The main hardware requirement for the development process will be a relatively new PC or laptop. The machine should be equipped with the following features: decent processor (Intel Core i5 or AMD Ryzen 5, or better), at least 8GB of RAM (preferably 16GB to work comfortably with development tools), as well as enough hard drive space (at least 256GB SSD). Software requirements for the development process include a suitable operating system (Windows, Mac OS, or Linux), a Java Development Kit (JDK) version compatible with Spring Boot (e.g., JDK 11, 17, or 21), a development environment such as IntelliJ IDEA, Eclipse, or VS Code with Java plugins, a build tool such as Maven or Gradle (provided by Spring Initializr), a relational database management system (locally installed MySQL Community Server or PostgreSQL), a web browser for testing (Chrome or Firefox), and a version control software (Git).

Hardware: In the case of the deployment environment, a Cloud Virtual Private Server (VPS) would be used. The VPS should have specifications, which include at least 1-2 vCPUs, 2GB-4GB RAM, and enough SSD storage (e.g., 20GB-50GB) in order to support the application, database, as well as all files uploaded (e.g., payment proofs). The operating system of the server should normally be some kind of Linux OS (e.g., Ubuntu or CentOS). In terms of software for the deployment server, it would be required to have the Java Runtime Environment (JRE) compatible with the application, chosen relational database server (MySQL or PostgreSQL), web server

or application server, if the embedded server of Spring Boot is not used (though Spring Boot uses embedded servers like Tomcat, Jetty, Undertow). Also, the set of useful tools for server administration, monitoring and security (e.g., firewall, SSH) would be needed. Moreover, an SSL/TLS certificate would be required in order to use HTTPS.

3.5 Technology Used Description

AI-Muneer Online Portal's development with the use of the Waterfall model entails systematic inclusion of a number of technologies in each phase of the development process. During the phase of System & Software Design (PSM1), the architecture describes the interactions between those technologies. For instance, the backend built on Spring Boot (Java) based on MVC architecture is to host the core business logic. That includes the functionality for managing venues details, booking requests, confirmation of payments, feedback, reporting, and notifications.

At this stage the relational database MySQL/PostgreSQL will be created to hold all of the permanent data, such as user passwords (hashed), hall description, availability, pricing policies, inquiries, payment proofs and feedback entries. This structured system will allow the implementation of the data fetching process necessary for the administrator reporting capabilities, directly.

The user interface will be created as the View part of the MVC architecture. It will use HTML5, CSS3 and JavaScript and will render server-side with a templating engine (Thymeleaf for example) controlled by Spring Boot. This methodology makes an extremely tight application where the backend can directly control the UI, which is exactly what this project aims to do.

The view will be constructed within the Model-View-Controller design pattern as the User Interface. The View will employ HTML5, CSS3, and JavaScript languages, and will involve server-side rendering using a template engine like Thymeleaf controlled by the Spring Boot. Such an approach creates an application that is tightly coupled where the backend has control over the User Interface, something that matches the needs of this project very well.

During the implementation stage, there will be a use of Cloud VPS where the packaged application of Spring Boot together with the database will be hosted. HTTPS (SSL/TLS) communication protocol will be required to ensure that all data exchanged between client browsers and the server remains secure. Waterfall model ensures that each of these technological elements undergoes design, implementation, and testing process systematically and in order.

3.6 Chapter Summary

The purpose of this chapter is to give a complete description of the System Development Methodology that has been used in the Al-Muneer Online Portal. This chapter has emphasized the importance of the Waterfall model in the implementation of the system development methodology. The reasons why this methodology has been adopted in this project is due to the fact that it follows a sequential approach and this makes it suitable where there are clearly defined requirements and deliverables. All the various phases of this methodology have been discussed in details including Requirements Analysis and Definition up to Deployment and Maintenance, assigning particular tasks to PSM1 and PSM2. Visual Gantt Charts have been utilized for defining the timeframe of each phase. This chapter also described the hardware and software requirements of the development and deployment environment of the portal. It has been shown how the technologies that have been chosen such as Spring Boot, relational database, and web frontend technology will work in the development process according to the waterfall methodology.